

1. (Fall 2024, #8) We intend to find the area of the region bounded by the graph of $f(x) = \frac{1}{1+x}$, the x -axis and the lines $x = 0$ and $x = 0.1$. To that end, we approximate f near $x = 0$ by a Taylor polynomial. What should the degree of the polynomial be at least to achieve an accuracy of 0.01?

$$\int_0^{0.1} \frac{1}{1+x} dx$$

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-x)^n \quad \text{for } x \in [0, 0.1].$$

$$\begin{aligned} R_n(x) &= \sum_{k=n+1}^{\infty} (-x)^k \\ &= \frac{(-x)^{n+1}}{1+x} \end{aligned}$$

$$R_n(x) = \frac{f^{(n+1)}(z)}{(n+1)!}$$

$$z \in (0, x) .$$

$$\Rightarrow |E_n| = \int_0^{0.1} \frac{x^{n+1}}{1+x} dx \leq \int_0^{0.1} x^{n+1} dx = \frac{0.1^{n+2}}{n+2} \leq 0.01$$

$$\Rightarrow n \geq 0 . \quad \text{choose } \boxed{n=0} .$$

$$\text{and } \int_0^{0.1} f(x) dx \approx \int_0^{0.1} f(0) dx = 0.1$$

$$\text{Actually: } \int_0^{0.1} f(x) dx = \int_0^{0.1} \frac{1}{1+x} dx = \ln(1+x) \Big|_0^{0.1} = \ln(1.1) \approx 0.0953 .$$

$$|0.1 - 0.0953| < 0.01 .$$

2. (Spring 2024, #9) Let $T_3(x)$ denote the Taylor polynomial of degree at most ~~3~~ about $x = \pi/4$ of $f(x) = \tan x$. Notice the first few derivatives of f are:

$$\begin{aligned} f(x) &= \tan x \\ f'(x) &= \sec^2 x = 1 + \tan^2 x \\ f''(x) &= 2 \tan x + 2 \tan^3 x \\ f'''(x) &= 6 \tan^4 x + 8 \tan^2 x + 2 \\ f^{(4)}(x) &= 24 \tan^5 x + 40 \tan^3 x + 16 \tan x \end{aligned}$$

$$\begin{aligned} \tan \frac{\pi}{4} &= 1 \\ \sec \frac{\pi}{4} &= \sqrt{2} \end{aligned}$$

You do not need to show these.

- (a) Find $T_3(x)$.

$$\begin{aligned} T_3(x) &= f\left(\frac{\pi}{4}\right) + \frac{f'\left(\frac{\pi}{4}\right)}{1!} \left(x - \frac{\pi}{4}\right) + \frac{f''\left(\frac{\pi}{4}\right)}{2!} \left(x - \frac{\pi}{4}\right)^2 + \frac{f'''\left(\frac{\pi}{4}\right)}{3!} \left(x - \frac{\pi}{4}\right)^3 \\ &= 1 + \frac{2}{1} \left(x - \frac{\pi}{4}\right) + \frac{2+2}{2} \left(x - \frac{\pi}{4}\right)^2 + \frac{6+8+2}{6} \left(x - \frac{\pi}{4}\right)^3 \\ &= 1 + 2 \left(x - \frac{\pi}{4}\right) + 2 \left(x - \frac{\pi}{4}\right)^2 + \frac{8}{3} \left(x - \frac{\pi}{4}\right)^3. \end{aligned}$$

- (b) Find a bound on the error when $T_3(x)$ is used to approximate $\tan(x)$ for $0 \leq x \leq \pi/4$.

$$R_3(x) = \frac{f^{(4)}(z)}{4!} \left(x - \frac{\pi}{4}\right)^4, \quad \text{for } z \in \left[x, \frac{\pi}{4}\right],$$

$$0 \leq |R_3(x)| \leq \frac{\left(x - \frac{\pi}{4}\right)^4}{4!} \cdot |24 + 40 + 16|.$$

$$= \frac{80}{24} \cdot \left(x - \frac{\pi}{4}\right)^4,$$

$$\frac{10}{3}$$

$$\text{for } x \in \left[0, \frac{\pi}{4}\right]: |R_3(x)| \leq \frac{10}{3} \cdot \left(\frac{\pi}{4}\right)^4.$$

3. (Fall 2023, #9) Let $f(x) = x^2 \ln x$.

(a) Compute $T_3(x)$ at 1.

$$\begin{aligned} T_3(x) &= f(1) + \frac{f'(1)}{1!}(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \frac{f'''(1)}{3!}(x-1)^3 \\ &= 0 + \frac{1}{1!}(x-1) + \frac{3}{2!}(x-1)^2 + \frac{2}{3!}(x-1)^3 \\ &= (x-1) + \frac{3}{2}(x-1)^2 + \frac{1}{3}(x-1)^3. \end{aligned}$$

$$f(x) = x^2 \ln x$$

$$f'(x) = 2x \ln x + x^2 \cdot \frac{1}{x} = \underbrace{2x \ln x} + x.$$

$$f''(x) = 2 \ln x + 2x \cdot \frac{1}{x} + 1 = 2 \ln x + 3.$$

$$f'''(x) = \frac{2}{x}$$

$$f^{(4)}(x) = \frac{-2}{x^2}.$$

(b) What is the maximum possible error in the approximation $f(x) \sim T_3(x)$, when $1/2 \leq x \leq 3/2$.

$$0 \leq |R_3(x)| \leq |x-1|^4 \cdot \frac{|f^{(4)}(z)|}{4!}$$

z between x and 1 . $x \in [\frac{1}{2}, \frac{3}{2}]$,

$$= |x-1|^4 \cdot \frac{2}{|z|^2 \cdot 4!}$$

$$\leq \left|\frac{1}{2} - 1\right|^4 \cdot \frac{2}{\left|\frac{1}{2}\right|^2 \cdot 4!} = \frac{1}{2^4} \cdot \frac{2}{\frac{1}{2^2} \cdot 24} = \frac{1}{48}.$$

4. (Spring 2023, #8) Consider the function $g(x) = (x-1)\ln x$.

(a) Find the degree-2 Taylor polynomial, $T_2(x)$, of g centered at $a = 1$.

$$g(x) = (x-1)\ln x$$

$$g'(x) = \ln x + \frac{x-1}{x} = \ln x + 1 - \frac{1}{x}$$

$$g''(x) = \frac{1}{x} + \frac{1}{x^2}$$

$$g'''(x) = -\frac{1}{x^2} - \frac{2}{x^3}$$

$$T_2(x) = g(1) + g'(1)(x-1) + \frac{g''(1)}{2}(x-1)^2$$

$$= 0 + 0 \cdot (x-1) + \frac{2}{2}(x-1)^2 = (x-1)^2$$

(b) Use $T_2(x)$ to approximate $g(1.5)$.

$$g(1.5) \approx T_2(1.5) = (1.5-1)^2 = \frac{1}{4}$$

(c) Estimate the accuracy of the above approximation.

$$R_2(x) = \frac{g'''(z)}{3!} (x-1)^3, \quad z \text{ between } x \text{ and } 1$$

$$\Rightarrow |R_2(x)| \leq \frac{\left|\frac{1}{z^2} + \frac{2}{z^3}\right|}{6} |x-1|^3$$

$$\Rightarrow |R_2(1.5)| \leq \frac{\left|\frac{1}{1} + \frac{2}{1}\right|}{6} \cdot |1.5-1|^3 = \frac{1}{2} \cdot 0.5^3 = \frac{1}{16}$$

5. (Fall 2021, #8) Let $f(x) = x^2 \ln(x+1)$. Find the 2nd-order Taylor polynomial of $f(x)$ centered at $x = 1$.

$$f(x) = x^2 \ln(x+1) \quad \Rightarrow \quad f(1) = \ln 2$$

$$f'(x) = 2x \ln(x+1) + \frac{x^2}{x+1} \quad \Rightarrow \quad f'(1) = 2 \ln 2 + \frac{1}{2}$$

$$f''(x) = 2 \ln(x+1) + \frac{2x}{x+1} + \frac{2x(x+1) - x^2}{(x+1)^2}$$

$$\Rightarrow f''(1) = 2 \ln 2 + \frac{2}{2} + \frac{2 \cdot 2 - 1}{2^2} = 2 \ln 2 + 1 + \frac{3}{4}$$

$$\Rightarrow T_2(x) = f(1) + f'(1)(x-1) + \frac{f''(1)}{2}(x-1)^2$$

$$= \ln 2 + (2 \ln 2 + \frac{1}{2})(x-1) + \frac{(2 \ln 2 + 1 + \frac{3}{4})}{2}(x-1)^2,$$